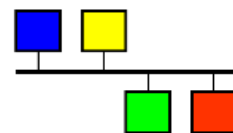


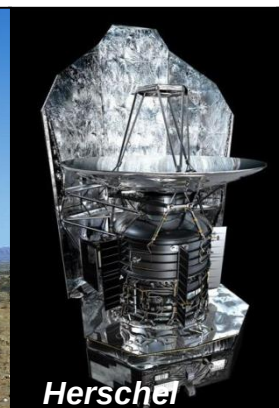
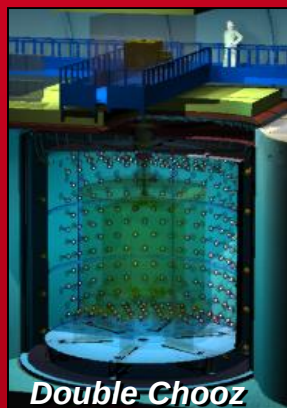
DE LA RECHERCHE À L'INDUSTRIE

cea

EPICS



WEB SERVICES USAGE

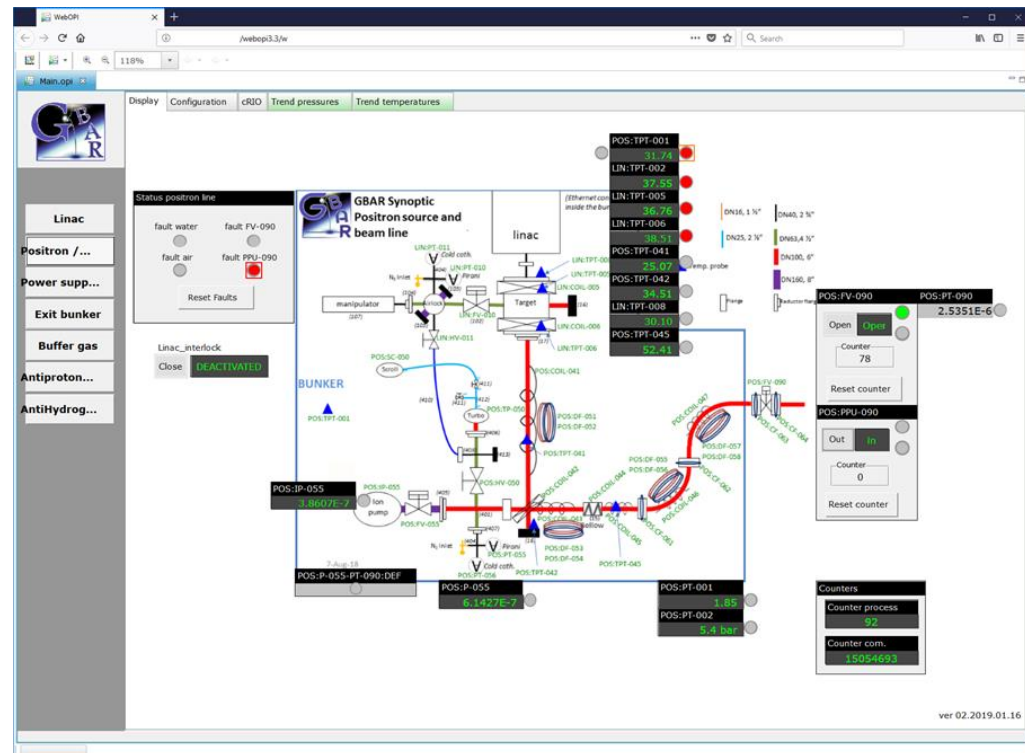


Paul LOTRUS & Katy SAINTIN

June 2019

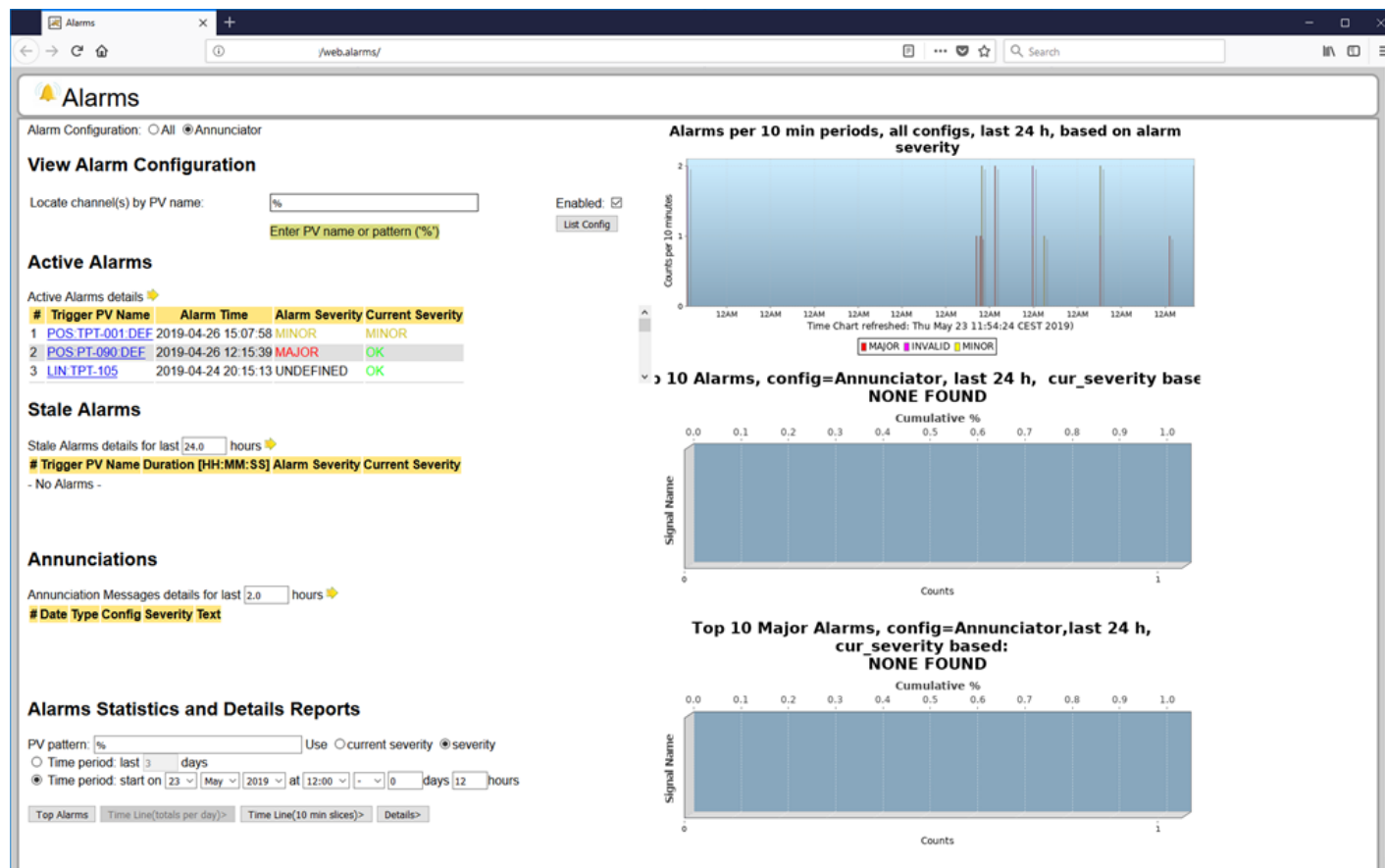
WEB OPI (SNS): GBAR experiment

- CSS RAP <https://github.com/CONTROLSYSTEMSTUDIO/CS-STUDIO/WIKI/WEBOP> running on Tomcat server gives easy access to online experiment status
- For read-only rights, we are using ca-gateway
- Performances: loading view is not fast
- Memory: Seems to have memory trouble, reboot of the application
- Resize of the controls is not an easy parameter for the developer
- Not CSS BOB compatible



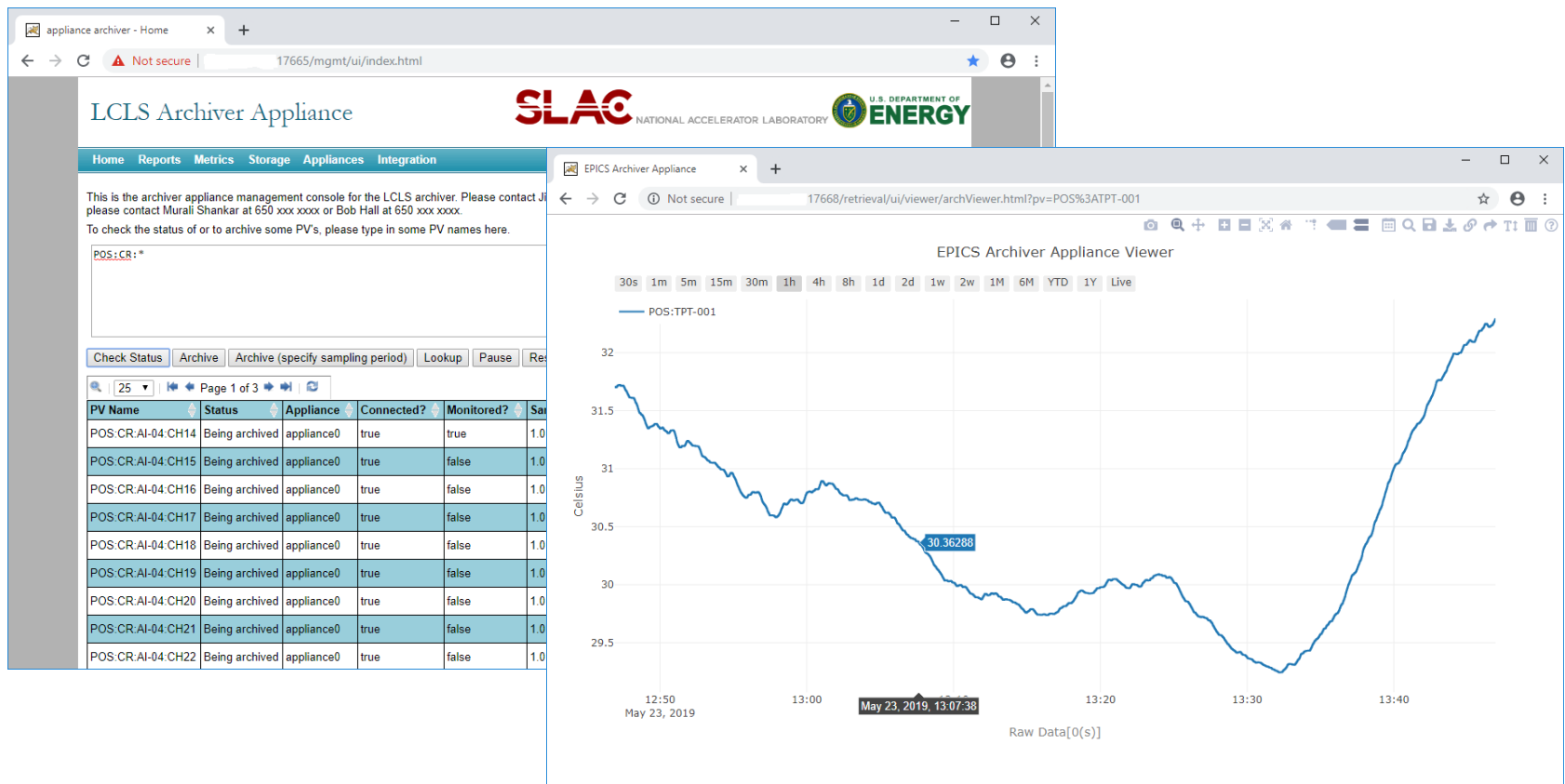
Website alarm (SNS): Test stand CEA and GBAR experiment

- We don't have a great feedback from the users, probably because it is not integrated with the synoptic view website
- Installation documentation is light, and so the configuration is not that easy



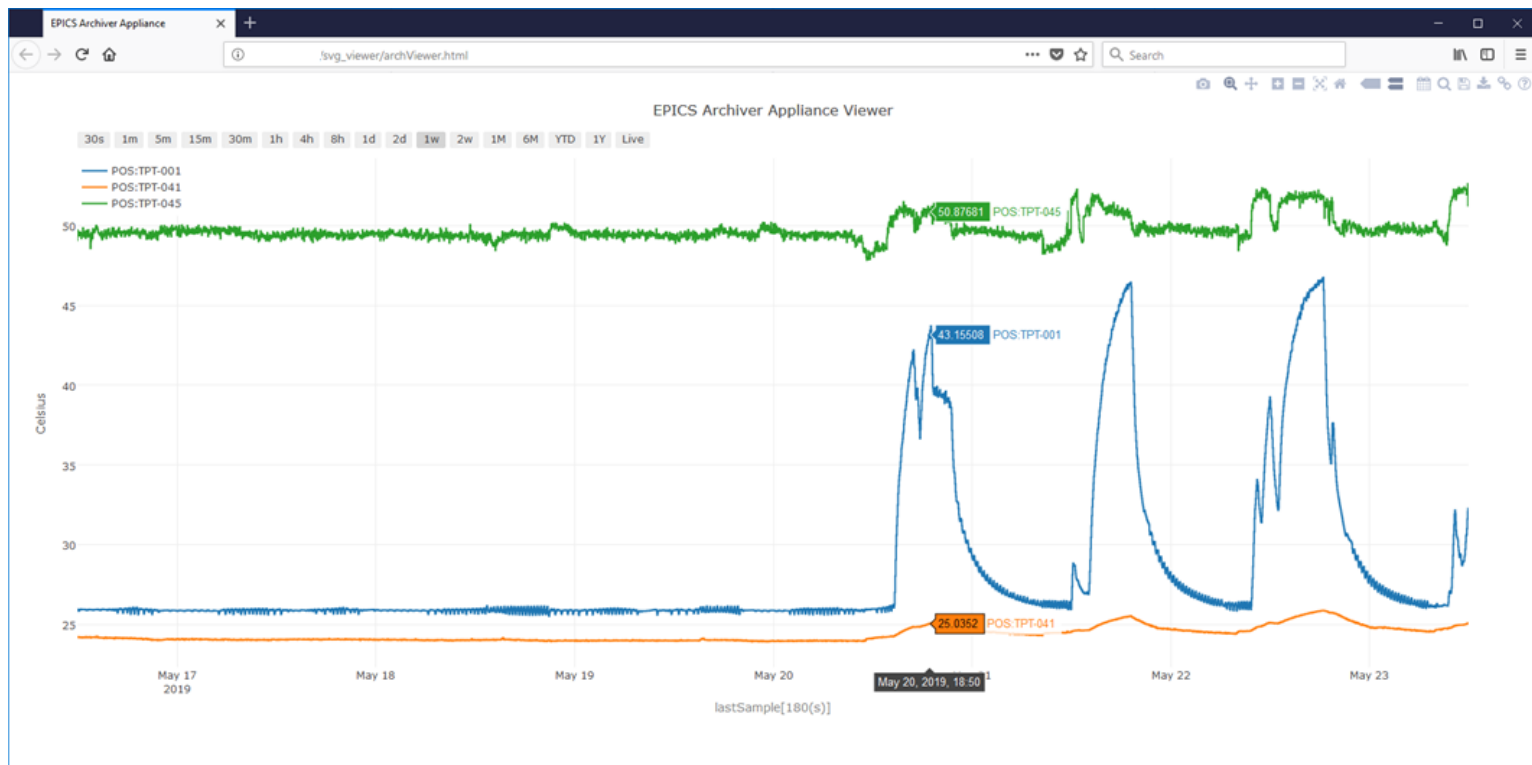
Archiver Appliance (SLAC): CEA Test stand and GBAR experiment

- Configuration and visualization with only one archiving machine
- Good documentation
- Good performance



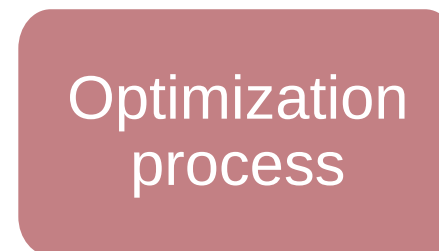
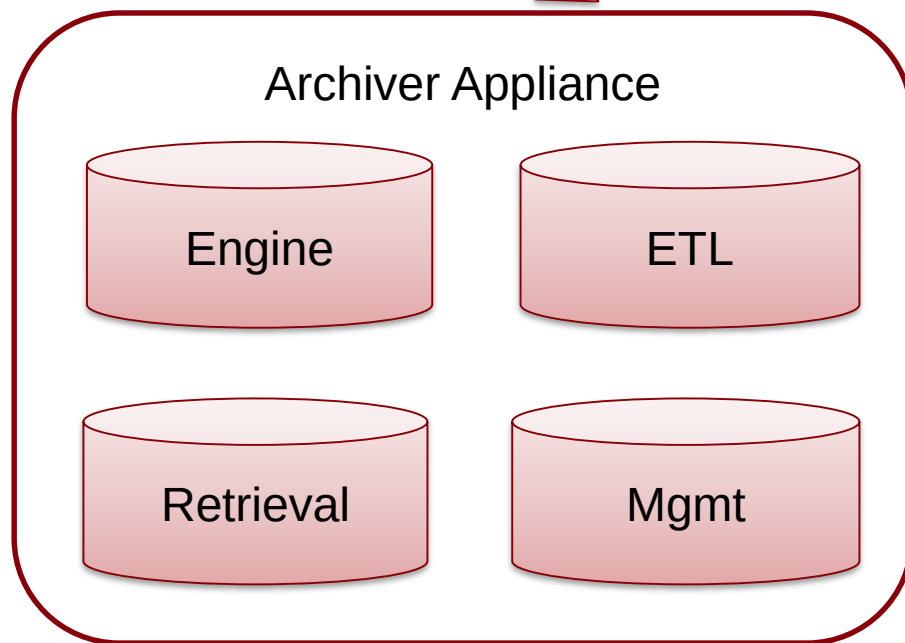
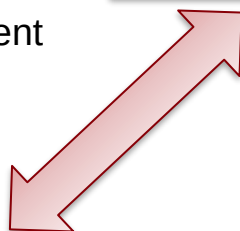
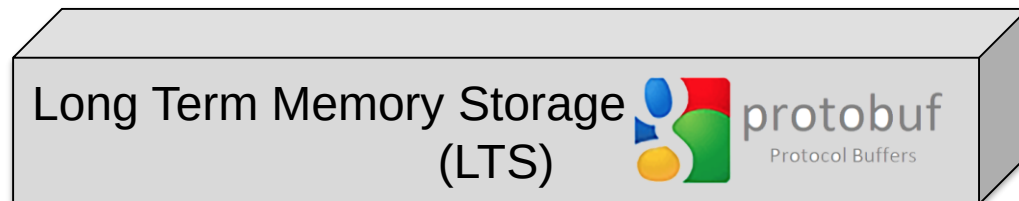
Archiver Appliance (SLAC): CEA Test stand and GBAR experiment

- Website running on tomcat server
- Only visualization
- Performance are good, good feedback from users
- It will be great to be able to get data with JSON request from this website. It could be used as a gateway for offline treatment



Small PC configuration: adapting Archive appliance

- Remove not interesting data
- Remove old data
- Not deployed for the moment



Memory Optimizer ON Archiver Appliance (MOONARCH) - Mozilla Firefox@irfulxis229

Memory Optimizer O... x

localhost:17669/moonarch/

MOON Archiver Appliance : Configuration page

User Configuration

Data depth: 1 S

Max Storage Duration: 5 year(s)

Optimization Mode: by File

PVs To Exclude: iphi:lebt:Thall
*BOB1_3_ON
ex : project.group:pvname ; can use wildcards (*)

Set Expert's Configuration

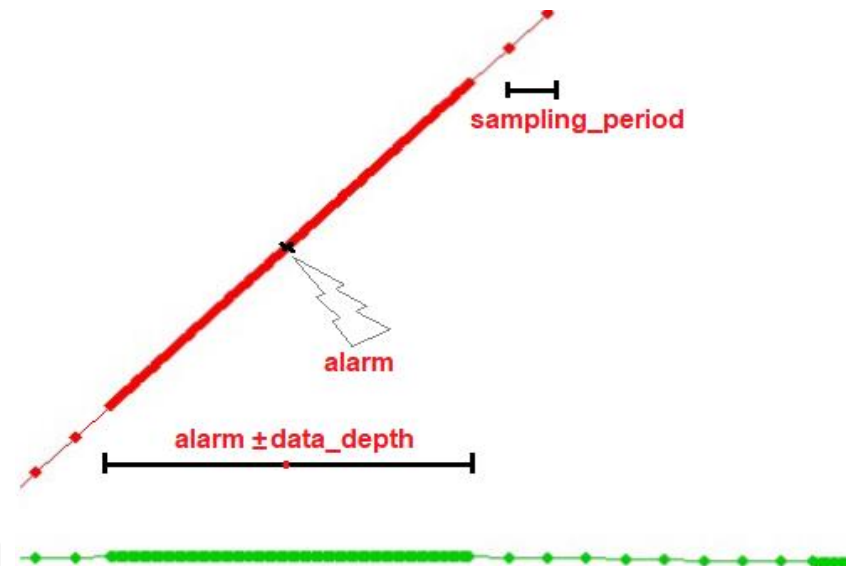
Load Default Configuration Leave Changes Change Configuration

Start Optimization

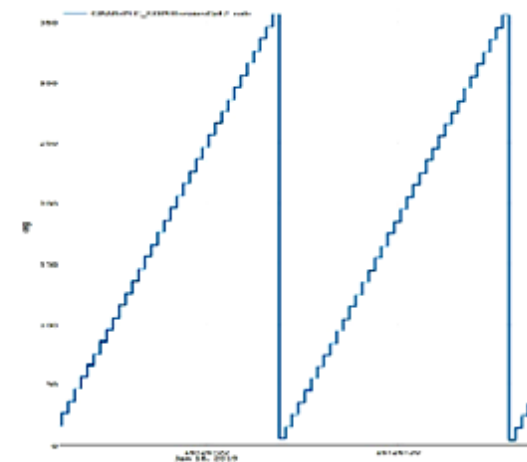
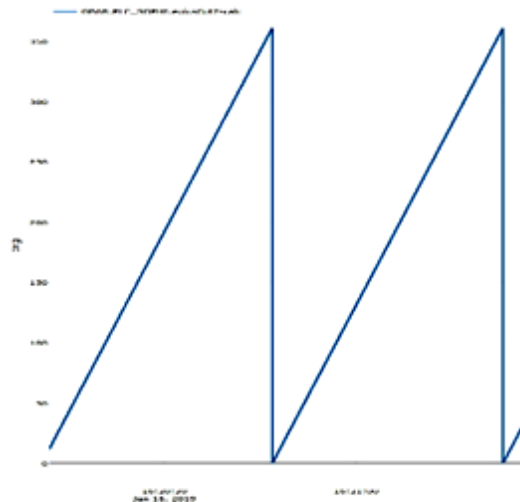
Stop Optimization

Parameters :

- Data depth
- Max storage duration
- Process variables to exclude
- Process period
- Optimization mode
- Process variables per task ...

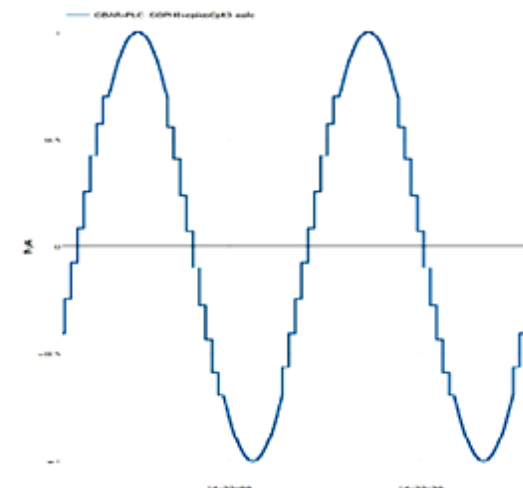
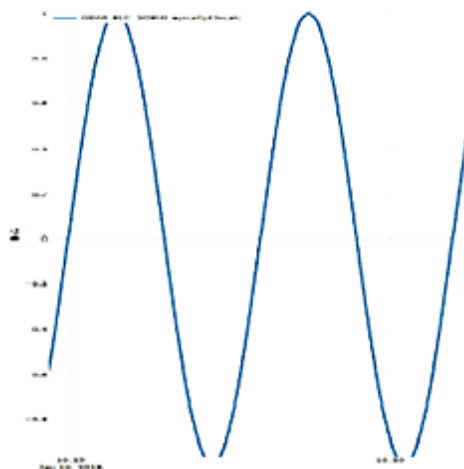


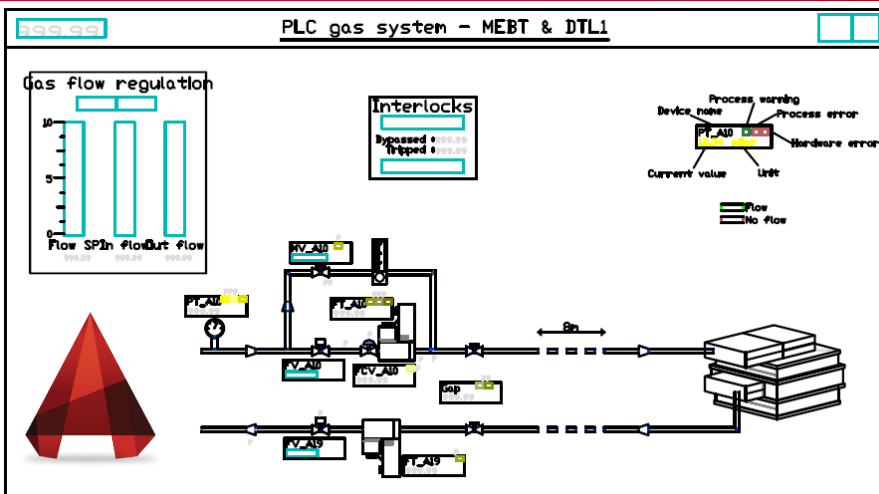
Type :	Dossier de fichiers
Emplacement :	D:\NEPICS\ArchivingData
Taille :	<u>1,32 Go (1 420 253 245 octets)</u>
Taille sur le disque :	1,32 Go (1 420 967 936 octets)
Contenu :	394 Fichiers, 6 Dossiers



Memory size - 40 %

Type :	Dossier de fichiers
Emplacement :	D:\EPICS\optimizedLts
Taille :	<u>799 Mo (837 895 889 octets)</u>
Taille sur le disque :	799 Mo (838 598 656 octets)
Contenu :	386 Fichiers, 6 Dossiers





Web EPICS

Apache Tomcat

WEB EPICS

localhost:8080/IOCviewer/webepics

COUNTER PV	10
DOUBLE PV	40.0
ENUM PV	three
BOOLEAN PV	ON
STRING PV	TEST STRING

MEBT & DTL1 line

Settings

Setpoint

Measure

Output

Low limit

High limit

Process warning

Process Error

Device name

Current value

Unit

Hardware error

FT_A10

PT_A10

MV_A10

FCV_A10

FT_A19

PT_A19

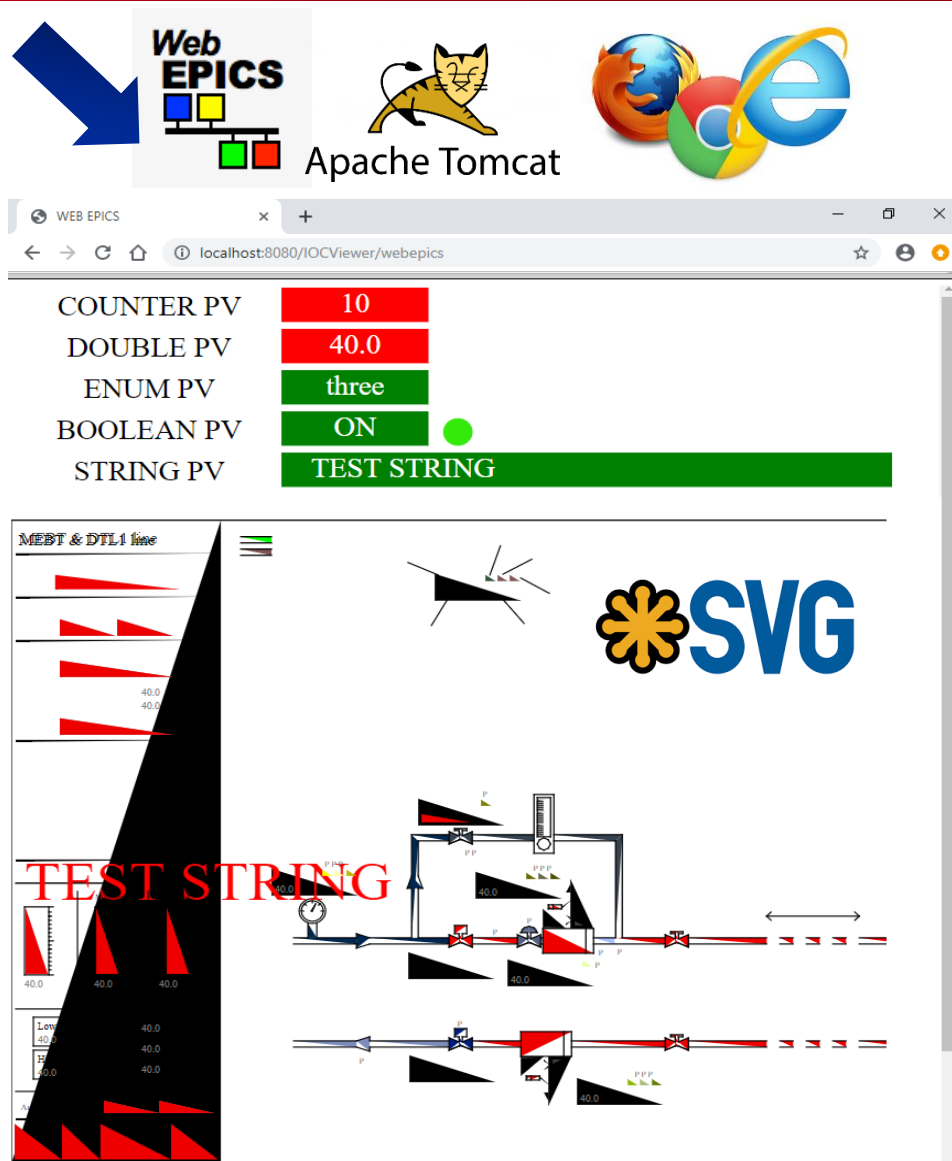
FCV_A19

Gap

8m

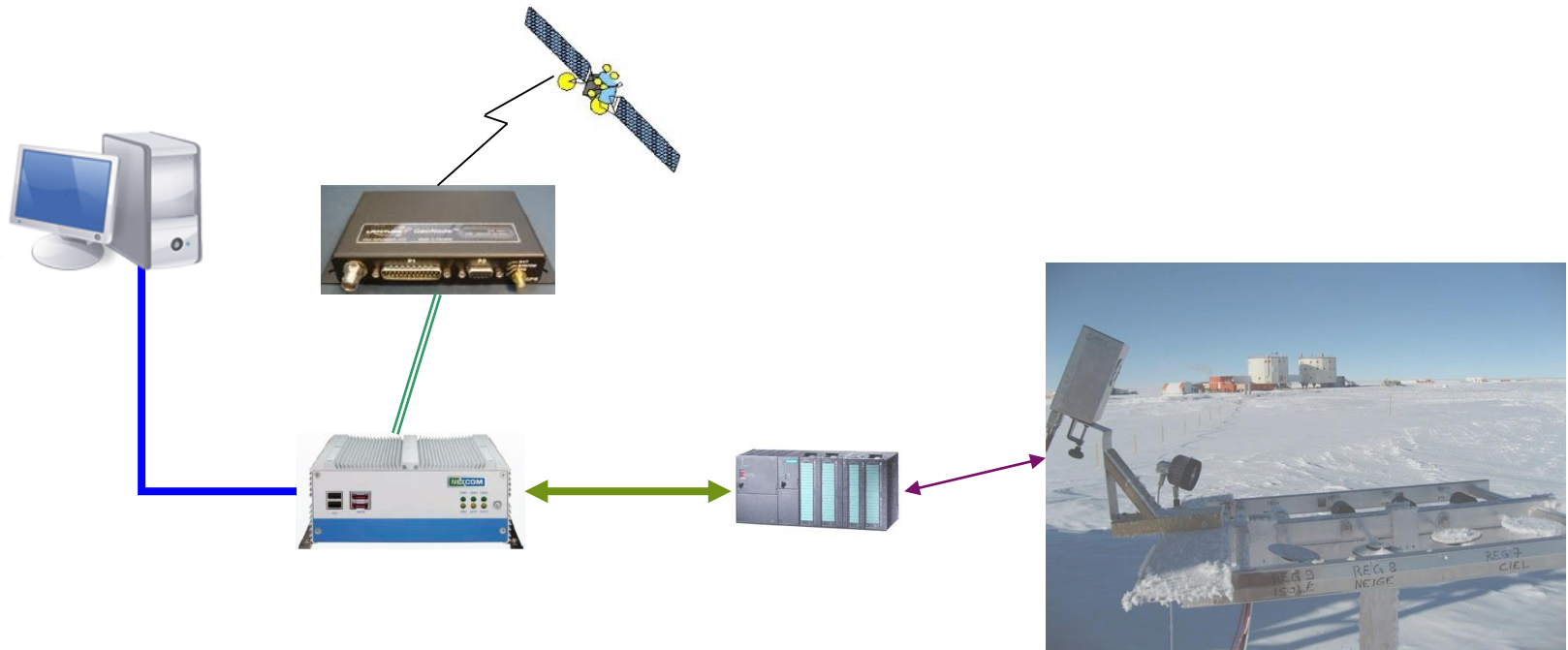
TEST STRING

SVG



Needs

- Synoptic website compatible with CSS .bob
- Website that integrates synoptic website, alarm and archiving data (trend)
- Light solution that can run on a small PC configuration
 - We deploy also control software on small experiments in different places in the world
 - Constraints: cost, space in the rack, environment condition
 - Example: Concordia on South pole: Experiment to test the environment for a future telescope: understand freeze on different materials, how to remove the snow, etc.



Thanks a lot to the developers and the facilities that support them



memegenerator.net